In Bohr-Sommerfeld theory of the atom electron orbit can be circular or elliptical. Gears are sometimes (for particular need) made elliptical in shape. (Fig. 5.52)



Fig. 5.52

The shape of our mother Earth is an oblate spheroid i.e., the solid of revolution of an ellipse about its minor axis, bulged along equatorial region and flat along the polar region.

The property of ellipse, any ray of light or sound released from a focus of the ellipse on touching the ellipse gets reflected to reach the other focus (Fig. 5.62), which could be proved using concepts of **incident rays** and **reflected rays** in Physics.

An exciting medical application of an ellipsoidal reflectors is a device called a **Lithotripter** (Fig. 5.4 and 5.63) that uses electromagnetic technology or ultrasound to generate a shock wave to pulverize kidney stones. The wave originates at one focus of the cross-sectional ellipse and is reflected to the kidney stone, which is positioned at the other focus. Recovery time following the use of this technique is much shorter than the conventional surgery, non-invasive and the mortality rate is lower.

5.7.3 Hyperbola

Some Comets travel in hyperbolic paths with the Sun at one focus, such comets pass by the Sun only one time unlike those in elliptical orbits, which reappear at intervals.

We also see hyperbolas in architecture, such as Mumbai Airport terminal (Fig. 5.53), in cross section of a planetarium, an locating ships (Fig. 5.54), or a cooling tower for a steam or nuclear power plant. (Fig. 5.5)



Fig. 5.53

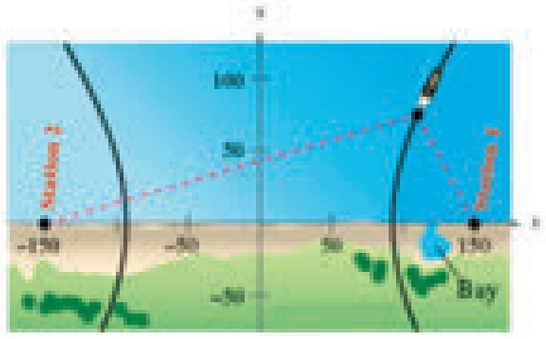


Fig. 5.54

Example 5.31

A semielliptical archway over a one-way road has a height of $3m$ and a width of $12m$. The truck has a width of $3m$ and a height of $2.7m$. Will the truck clear the opening of the archway? (Fig. 5.6)

Solution

Since the truck's width is $3m$, to determine the clearance, we must find the height of the archway $1.5m$ from the centre. If this height is $2.7m$ or less the truck will not clear the archway.

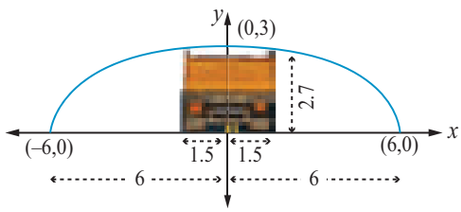


Fig. 5.55

From the diagram $a = 6$ and $b = 3$ yielding the equation of ellipse as $\frac{x^2}{6^2} + \frac{y^2}{3^2} = 1$.

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